Laura S. Bumbulis

Department of Statistics & Actuarial Science
University of Waterloo
Waterloo, ON, Canada, N2L 3G1

lbumbulis.github.io
lsbumbul@uwaterloo.ca
ORCiD

Education, University of Waterloo

PhD Statistics 2024–present
Thesis: *Design and Analysis of Studies Involving Incompletely Observed Multistate Processes*
Advisor: Richard J. Cook
Overall Average (6 courses): 96%

MMath Statistics 2023–2024
Masters Research Paper: *Robustness and Efficiency Considerations when Testing Process Reliability with a Limit of Detection*
Advisor: Richard J. Cook
Overall Average (7 courses): 97%

BMath (Co-op) 2017–2022
Majors: Statistics; Combinatorics & Optimization
Minor: Pure Mathematics
Overall Average (45 courses): 94%

Publications

5. Eder, L., Shahab, S., Gillespie, S., **Bumbulis, L.**, Emanoilidis, H., Compher, C., Scher, J., Gladman, D., Cook, R., Chandran, V., & Ogdie, A. (2026). Dietary interventions in psoriatic arthritis: a randomized controlled clinical trial. *The Journal of Rheumatology*, *53* (Suppl 1), 138. DOI:10.3899/jrheum.2026-0447.177
4. Cook, R. J., **Bumbulis, L. S.**, Kim, S. D., & Albert, P. S. (2026). Modeling the effects of smoking on lung cancer. *Observational Studies*, *12* (2), 193–209. muse.jhu.edu/pub/56/article/999745
3. Almansouri, A. Y., Li, J., **Bumbulis, L.**, Alhadari, A., Colaco, K., Harvey, P., Akhtari, S., Chandran, V., Gladman, D. D., Cook, R. J., & Eder, L. (2026). Incidence rate and risk factors of arrhythmias in patients with psoriatic arthritis. *RMD Open*, *12* (1), e005512. DOI:10.1136/rmdopen-2025-005512
2. Colaco, K., **Bumbulis, L.**, Chandran, V., Cook, R., Gladman, D., & Eder, L. (2025). The effect of biologic therapies on serum metabolic biomarkers in patients with psoriatic arthritis. *The Journal of Rheumatology*, *52* (Suppl 2), 27–28. DOI:10.3899/jrheum.2025-0314.TOUR2C
1. **Bumbulis, L. S.**, & Cook, R. J. (2025). Robustness and efficiency considerations when testing process reliability with a limit of detection. *Mathematics*, *13* (8), 1274. DOI:10.3390/math13081274

External Research Funding

NSERC Canada Graduate Scholarship – Doctoral	2025–2028	$ 120,000
Ontario Graduate Scholarship (OGS) / QEII-GSST (Declined)	2025–2026	$ 15,000

Scholarships & Awards

Conference Presentation Awards

SSC Student Research Presentation Award: Biostatistics Competition	2025	$ 500
WSSAF Presentation Award	2024	$ 500

University of Waterloo Scholarships & Awards

A list of acronyms can be found on page 4.

Doctoral

President's Graduate Scholarship	2025–2028	$30,000
Ontario Graduate Fellowship	2026	$ 4,000
University of Waterloo Graduate Scholarship (×2)	2025–2026	$ 2,000
SAS Chair's Graduate Award (×2)	2025–2026	$ 1,500
SAS Department Comprehensive Examination Scholarship	2025	$ 1,000
Russell Hiscock Graduate Scholarship in SAS	2025	$ 2,000
SAS Department Teaching Assistant Award	2024	$ 500
Cotton Family Women in Mathematics Graduate Scholarship	2024	$ 3,000
Provost Doctoral Entrance Award for Women	2024	$ 5,000

Masters

SAS Department Outstanding Academic Performance Award	2024	$ 500
University of Waterloo Graduate Scholarship (×2)	2024	$ 3,000
D.A. Sprott Entrance Award	2023	$ 1,000

Undergraduate

President's International Experience Award	2022	$ 1,500
Faculty of Mathematics Scholarship	2017–2022	$10,000
Co-op Student of the Year, Faculty of Mathematics	2021	
U Sports Academic All-Canadian	2018–2021	
President's Research Award	2018	$ 1,500
President's Scholarship of Distinction	2017	$ 2,000

Presentations

Contributed Conference Presentations

6. *Multistate Joint Modelling of Marker and Failure Data under Dual Censoring Schemes.* ISCB Annual Conference, Freiburg (Germany). September 2026.
5. *Challenges and Opportunities in the Analysis of Cervical Screening Data.* SSC Annual Meeting, Hamilton, ON. June 2026.
4. *Joint Models for Marker and Failure Data under Dual Censoring Schemes.* ENAR Spring Meeting, Indianapolis, IN (USA). March 2026.
3. *Dual Censoring Schemes for Marker and Failure Time Processes.* SSC Annual Meeting, Saskatoon, SK. May 2025.
2. *Robustness and Efficiency Considerations when Testing Process Reliability with a Limit of Detection.* WSSAF, Waterloo, ON. October 2024.
1. *Testing Process Reliability under a Limit of Detection: Issues of Robustness and Efficiency.* SSC Annual Meeting, St. John's, NL. June 2024.

Teaching Experience

Guest Lectures & Exam Reviews

STAT 431: Generalized Linear Models — Spring 2026
Guest lecture on overdispersion (80 mins)
Instructor: Cecilia A. Cotton

STAT 450: Estimation and Hypothesis Testing — Winter 2025
Guest lecture on Bayesian estimation (50 mins)
Instructor: Peter W. MacDonald

STAT 850: Estimation and Hypothesis Testing (Graduate-Level) — Fall 2024
Two final exam review sessions (3 hours each)
Instructor: Yingli Qin

Teaching Assistantships

STAT 231: Statistics — Spring 2025

STAT 230: Probability (Online) Winter 2025
STAT 450: Estimation and Hypothesis Testing Winter 2025
STAT 330: Mathematical Statistics † Fall 2024

Credentials

Foundations for University Teaching in SAS Certificate Winter 2025

† *Received SAS Department TA award.*

All teaching experience affiliated with the University of Waterloo; see course descriptions here.

Professional Activities

External Roles

Member of CANSSI Ontario Student PhD and Research Network (SPARK) 2025–2026
Organizer of 2026 CANSSI Ontario STatistics (CAST) seminar by Kuan Liu

Internal Roles

Co-organizer for SAS Student Seminar Series 2025–present
Graduate representative for SAS Department (elected) 2025–2027
Co-Chair of 6th Waterloo Student Conference in SAS & Finance 2026
Graduate representative for Women in Mathematics Committee 2024–2025
Co-organizer for 2025 Navigating Your Careers Conference, University of Waterloo
Co-organizer for 2025 Ontario Graduate Mathematics Conference, University of Toronto

Conference Session Chair

Invited Sessions

SSC Annual Meeting: *Design & Analysis of Two-Phase Studies* 2027

Contributed Sessions

ENAR Spring Meeting: *Artificial Intelligence for Biomedical Decision-Making* 2026
WSSAF: *Methods for Inference and Improving Efficiency* 2024

Work Experience

Data Analyst / Software Developer (Full-time), Emmetros Jan.–Apr. 2023
Data Scientist (Co-op), Omnium Jan.–Apr. 2021, May–Aug. 2022
Nominated for 2022 CEIA Student Achievement Award
Bioinformatics Analyst (Co-op), Ontario Institute for Cancer Research May–Aug. 2021
Awarded 2021 Co-op Student of the Year, Faculty of Mathematics
Data Analyst / Software Developer (Co-op/Part-time), Emmetros Sep. 2019–Dec. 2020
Nominated for 2020 Co-op Student of the Year, Faculty of Mathematics

Additional Activities

Intramural Sports, University of Waterloo

7v7 Soccer Spring 2026
Volleyball (Team Captain) Winter 2025
Flag Football Spring 2024

Varsity Swim Team, University of Waterloo

Volunteer Assistant Coach 2021–2022
Co-Captain 2020–2021
Member 2017–2021

Music

Piano Player 2007–present
RCM Level 9 Harmony Certificate 2013
RCM Level 7 Piano Certificate 2013

Acronyms

ENAR	Eastern North American Region (International Biometric Society)
CANSSI	Canadian Statistical Sciences Institute
CEIA	Canadian Education & Internship Association
ISCB	International Society for Clinical Biostatistics
LiDS	Lifetime Data Science
NSERC	Natural Sciences & Engineering Research Council
QEII-GSST	Queen Elizabeth II Graduate Scholarship in Science & Technology
RCM	Royal Conservatory of Music
RMD	Rheumatic & Musculoskeletal Diseases
SAS	Statistics & Actuarial Science
SSC	Statistical Society of Canada
WSSAF	Waterloo Student Conference in Statistics, Actuarial Science, & Finance